

# LAST TUTORIAL

ECON 101–INTRODUCTION TO MICROECONOMICS

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Please discuss the following exercises. Exercise 2 should be covered in full, so please start with that one.

1. Please discuss the difference between the short-run and long-run costs. As an example, calculate the short-run and long-run costs schedules for the following production function

$$q = AL^{1/3}K^{1/3}.$$

2. Suppose that an economy has two individuals, A and B. A's demand curve is given by  $q_D^A = 100,000 - 2p$  and B's demand curve is given by  $q_D^B = 150,000 - 6p$ . Furthermore, there are initially 10,000 identical firms with the following cost structure: Total Cost:  $TC(q) = 10,000 + 100q + 10q^2$ ,  
Marginal Cost:  $MC(q) = 100 + 20q$ .

- #1 Find the market demand curve. Provide an expression for  $Q_D$
- #2 What is the firm's shutdown point and what is the firm's supply schedule?
- #3 What is the market price and quantity?
- #4 Will firms enter or exit?
- #5 Calculate the long-run equilibrium price, quantity, and number of firms.